

5G RAN

Multi-Frequency Smart Aggregation Feature Parameter Description

Issue	Draft A
Date	2025-12-31



Copyright © Huawei Technologies Co., Ltd. 2026. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Huawei Technologies Co., Ltd.

Trademarks and Permissions



HUAWEI and other Huawei trademarks are trademarks of Huawei Technologies Co., Ltd.

All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The purchased products, services and features are stipulated by the contract made between Huawei and the customer. All or part of the products, services and features described in this document may not be within the purchase scope or the usage scope. Unless otherwise specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without warranties, guarantees or representations of any kind, either express or implied.

The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:

<https://www.huawei.com/en/psirt/vul-response-process>

For vulnerability information, enterprise customers can visit the following web page:

<https://securitybulletin.huawei.com/enterprise/en/security-advisory>

Contents

1 Change History.....**1**

2 About This Document.....**3**

2.1 General Statements..... 3

2.2 Features in This Document..... 4

2.3 Differences..... 5

3 Overview.....**6**

4 Inter-Carrier Frequency Selective Scheduling.....**9**

4.1 Principles..... 9

4.2 Network Analysis..... 10

4.2.1 Benefits..... 10

4.2.2 Impacts..... 10

4.3 Requirements..... 11

4.3.1 Licenses..... 11

4.3.2 Functions..... 11

4.3.2.1 Prerequisite Functions..... 12

4.3.2.2 Mutually Exclusive Functions..... 13

4.3.2.3 Function Impacts..... 13

4.3.3 Hardware..... 14

4.3.4 Networking..... 15

4.3.5 Others..... 15

4.4 Operation and Maintenance..... 15

4.4.1 Data Configuration..... 15

4.4.1.1 Data Preparation..... 15

4.4.1.2 Using MML Commands..... 16

4.4.1.3 Using the MAE-Deployment..... 17

4.4.2 Activation Verification..... 17

4.4.3 Network Monitoring..... 17

5 Smart Inter-Frequency Channel Assistance (Trial).....**18**

5.1 CCE-based Data Split Stop for CA UEs..... 18

5.1.1 Principles..... 18

5.1.1.1 Triggering of CCE-based Data Split Stop for CA UEs..... 20

5.1.1.2 UE Selection..... 20

5.1.1.3 CCE-based Data Split Stop and Resumption for CA UEs.....21

5.1.2 Network Analysis..... 23

5.1.2.1 Benefits..... 23

5.1.2.2 Impacts..... 23

5.1.3 Requirements..... 23

5.1.3.1 Licenses..... 23

5.1.3.2 Functions..... 24

5.1.3.2.1 Prerequisite Functions..... 24

5.1.3.2.2 Mutually Exclusive Functions..... 25

5.1.3.2.3 Function Impacts..... 26

5.1.3.3 Hardware..... 30

5.1.3.4 Networking..... 30

5.1.4 Operation and Maintenance..... 30

5.1.4.1 Data Configuration..... 30

5.1.4.1.1 Data Preparation..... 30

5.1.4.1.2 Using MML Commands..... 31

5.1.4.1.3 Using the MAE-Deployment..... 32

5.1.4.2 Activation Verification..... 32

5.1.4.3 Network Monitoring..... 33

6 Multi-Frequency Joint Scheduling.....34

6.1 Principles..... 34

6.2 Network Analysis..... 36

6.2.1 Benefits..... 36

6.2.2 Impacts..... 36

6.3 Requirements..... 37

6.3.1 Licenses..... 37

6.3.2 Functions..... 38

6.3.2.1 Prerequisite Functions..... 38

6.3.2.2 Mutually Exclusive Functions..... 39

6.3.2.3 Function Impacts..... 42

6.3.3 Hardware..... 59

6.3.4 Networking..... 60

6.3.5 Cells..... 60

6.4 Operation and Maintenance..... 60

6.4.1 Data Configuration..... 60

6.4.1.1 Data Preparation..... 60

6.4.1.2 Using MML Commands..... 61

6.4.1.3 Using the MAE-Deployment..... 62

6.4.2 Activation Verification..... 62

6.4.3 Network Monitoring..... 62

7 Single DCI (Trial)..... 63

7.1 Principles..... 63

7.1.1 Triggering Conditions..... 65

7.1.2 Selecting Cells for Joint Scheduling..... 67

7.1.3 Performing Single-DCI-based Scheduling..... 67

7.2 Network Analysis..... 68

7.2.1 Benefits..... 68

7.2.2 Impacts..... 68

7.3 Requirements..... 68

7.3.1 Licenses..... 69

7.3.2 Functions..... 69

7.3.2.1 Prerequisite Functions..... 69

7.3.2.2 Mutually Exclusive Functions..... 70

7.3.2.3 Function Impacts..... 73

7.3.3 Hardware..... 82

7.3.4 Networking..... 82

7.3.5 Cells..... 82

7.3.6 Others..... 82

7.4 Operation and Maintenance..... 83

7.4.1 Data Configuration..... 83

7.4.1.1 Data Preparation..... 83

7.4.1.2 Using MML Commands..... 83

7.4.1.3 Using the MAE-Deployment..... 84

7.4.2 Activation Verification..... 84

7.4.3 Network Monitoring..... 84

8 Parameters..... 85

9 Counters..... 86

10 Glossary..... 87

11 Reference Documents..... 88

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 05 (2025-11-07).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between single DCI and time-domain power saving BWP. For details, see 7.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship between CA Performance Boost Based on Multi-dimensional Optimization and RF module fast deep dormancy. For details, see 6.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed. Examples of such changes include changes in function names.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial

features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-070201	Multi-Frequency Spatial Smart Carrier Selection	4 Inter-Carrier Frequency Selective Scheduling
FOFD-070214	Smart Inter-Frequency Channel Assistance (Trial)	5.1 CCE-based Data Split Stop for CA UEs
FOFD-081205	CA Performance Boost Based on Multi-dimensional Optimization	6 Multi-Frequency Joint Scheduling
FOFD-081234	Single DCI (Trial)	7 Single DCI (Trial)

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Inter-carrier frequency selective scheduling	None	4 Inter-Carrier Frequency Selective Scheduling
CCE-based data split stop for CA UEs	None	5.1 CCE-based Data Split Stop for CA UEs
Multi-frequency joint scheduling	None	6 Multi-Frequency Joint Scheduling
Single DCI	Supported only in SA networking	7 Single DCI (Trial)

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Inter-carrier frequency selective scheduling	Supported only in low frequency bands	4 Inter-Carrier Frequency Selective Scheduling
CCE-based data split stop for CA UEs	Supported only in low frequency bands	5.1 CCE-based Data Split Stop for CA UEs
Multi-frequency joint scheduling	Supported only in low frequency bands	6 Multi-Frequency Joint Scheduling
Single DCI	Supported only in low frequency bands	7 Single DCI (Trial)

3 Overview

iHashBand is a 5G intelligent coordination solution proposed by Huawei for multi-frequency networks. This solution is used to efficiently coordinate various resources (including spectrums, channels, beams, and base station models) of multi-frequency networks and to implement intelligent convergence and mutual assistance among frequency bands. By fully exploiting the characteristics and advantages of each frequency band, it can provide optimal network performance and user experience even with limited spectrum resources, thereby maximizing spectrum value. The iHashBand solution includes functions in the following categories:

- **Multi-frequency smart selection:** Functions in this category can intelligently select carriers or carrier combinations that provide the optimal user experience for UEs based on factors such as the spectral efficiency, bandwidth, and load, thereby improving user experience. For details about the related functions, see *Multi-Frequency Smart Selection*.
- **Multi-frequency smart aggregation:** Functions in this category can quickly and intelligently aggregate time-frequency resources of multiple cells operating on different frequencies for individual UEs, thereby increasing UE or cell throughput. For details about the related functions, see this document and *CA Experience Improvement*. Examples of the functions are as follows:
 - 3CC aggregation: This function aggregates three intra-band or inter-band carriers in the downlink to provide higher bandwidth. As a UE in the 3CC aggregation state can use three carriers at the same time, it can enjoy a higher peak data rate. For details, see *CA Experience Improvement*.
 - Inter-carrier frequency selective scheduling: This function enables the base station to allocate resources of the carriers with high spectral efficiency and low load to individual downlink CA UEs whenever possible based on the spectral efficiency difference and millisecond-level load difference among carriers. This achieves appropriate downlink resource allocation for CA UEs and thereby increases the average downlink UE throughput on the entire network.

The multi-frequency smart aggregation functions described in this document and their application scenarios are listed in [Table 3-1](#).

Table 3-1 Multi-frequency smart aggregation functions and their application scenarios

Feature	Function	Application Scenario	Chapter/ Section
FOFD-070201 Multi-Frequency Spatial Smart Carrier Selection	Inter-carrier frequency selective scheduling	For each UE in the downlink CA state, there are spectral efficiency differences and large millisecond-level load differences between carriers. When such a UE is scheduled, the inter-carrier frequency selective scheduling function enables appropriate downlink resource allocation for the UE. This results in an increase in the average downlink UE throughput on the entire network.	4 Inter-Carrier Frequency Selective Scheduling
FOFD-070214 Smart Inter-Frequency Channel Assistance (Trial)	CCE-based data split stop for CA UEs	In CA scenarios, if there are a large number of small-packet UEs at the NR cell edge, CCE resource consumption is high and CCE resources are likely to become insufficient, resulting in a negative impact on the user experience of most UEs. The CCE-based data split stop for CA UEs function enables the base station to select CA UEs for which high CCE aggregation levels are used in NR cells in the CCE resource congestion state and stop data distribution to the carriers in the congestion state for these UEs. This allows some CCE resources to be released, thereby increasing the average cell throughput.	5.1 CCE-based Data Split Stop for CA UEs

Feature	Function	Application Scenario	Chapter/ Section
FOFD-081205 CA Performance Boost Based on Multi- dimensional Optimization	Multi- frequency joint scheduling	In multi-carrier scenarios, the downlink scheduling of a CA UE in its PCell and SCells is relatively independent of each other. As a result, RB resources cannot precisely match traffic volumes. In addition, the traditional downlink scheduling solution considers only one of the time and frequency domains. To address this issue, the multi-frequency joint scheduling function is introduced. This function achieves precise scheduling for CA UEs and optimizes downlink scheduling in both the time and frequency domains, increasing the average downlink throughput of UEs.	6 Multi-Frequency Joint Scheduling
FOFD-081234 Single DCI (Trial)	Single DCI	In multi-carrier scenarios, when a CA UE is scheduled on more than one carrier, carrier-specific CCE resources are used for indicating PDSCH scheduling on each of the carriers involved. The PDSCH scheduling may fail due to insufficient PDCCH CCEs, negatively affecting the average downlink throughput of CA UEs. To address this issue, the single DCI function is introduced. This function achieves inter-carrier joint scheduling, which increases the average downlink throughput of UEs.	7 Single DCI (Trial)

 NOTE

- In NSA networking or NSA and SA hybrid networking, when the gNodeB selects carriers for UEs, it filters carrier combinations based on UE capabilities and LTE-NR frequency relationships. For details, see *EPC-based NSA Basics*.
- For basic functions of CA, see *Carrier Aggregation*.

4 Inter-Carrier Frequency Selective Scheduling

4.1 Principles

On a multi-carrier network, the spectral efficiencies of different frequencies concurrently measured by a CA UE vary due to propagation loss and multipath effect. In addition, there are load differences between carriers. Against this background, inter-carrier frequency selective scheduling is introduced. This function works in intra-base-station scenarios. With the inter-carrier frequency selective scheduling function, the base station allocates resources of carriers with high spectral efficiency and low load to individual downlink CA UEs whenever possible based on the differences in spectral efficiency and millisecond-level load between carriers. (For details about load statistics, see [Load Statistics](#).) This helps achieve appropriate downlink resource allocation for CA UEs and thereby increases the average downlink UE throughput on the entire network. How the function works is illustrated in [Figure 4-1](#).

Figure 4-1 Schematic diagram of inter-carrier frequency selective scheduling



Frequency 2 (F2) has higher spectral efficiency and lower load than F1.

Intra-base-station inter-carrier frequency selective scheduling is controlled by the **CROSS_CARRIER_FREQ_SEL_SCH_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter.

After inter-carrier frequency selective scheduling is enabled, it is recommended that RSRP threshold 2 for event A5 related to CA for the PCell be less than the RSRP threshold for event A2 related to coverage-based inter-frequency handovers for SCells and that the RSRP threshold for event A2 related to CA for the PCell be less than the RSRP threshold 2 for event A5 related to CA for the PCell, so that inter-carrier frequency selective scheduling can take effect for more UEs after SCells are activated for them. The thresholds are specified by different parameters as follows:

- RSRP threshold 2 for event A5 related to CA is specified by the **NRCellCaMgmtConfig.CaA5RsrpThld2** parameter.
- The RSRP threshold for event A2 related to coverage-based inter-frequency handovers is specified by the **NRCellInterFHoMeaGrp.CovInterFreqA2RsrpThld** parameter.
- The RSRP threshold for event A2 related to CA is specified by the **NRCellCaMgmtConfig.CaA2RsrpThld** parameter.

Load Statistics

The load statistics vary depending on the setting of the **AVG_LOAD_BASED_DATA_SPLIT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter.

- If this option is deselected, the average load of the PCell and SCells within a measurement period will not be considered.
- If this option is selected, the average load of the PCell and SCells within a measurement period will be considered. This statistics method is superior. Therefore, it is recommended that this option be selected when inter-carrier frequency selective scheduling is enabled.

4.2 Network Analysis

4.2.1 Benefits

It is recommended that inter-carrier frequency selective scheduling be enabled when the cell load is medium or heavy. After this function is enabled, the overall **User Downlink Average Throughput (DU)** on the network increases.

It is recommended that inter-carrier frequency selective scheduling be enabled for all cells in an area. Otherwise, it is possible that the maximum gains in average downlink UE throughput cannot be achieved.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

Inter-carrier frequency selective scheduling may have the following impacts on the network:

- The downlink PRB usage of the cell involved (indicated by **N.PRB.DL.Used.Avg** divided by **N.PRB.DL.Avail.Avg**) increases.

- The average downlink MCS index of the cell involved decreases.
- The CA- and non-CA-related traffic distributions in the cell involved change. As a result, the values of CA-related counters or indicators also change. For example, the values of **N.CA.PR.B.DL.PCell.Used.Avg** and **N.CA.PR.B.DL.SCell.Used.Avg** and the average throughput of CA UEs with SCells activated will change. The average throughput of CA UEs with SCells activated is the result of $(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot)/N.CA.ThpTime.DL.RmvLastSlot$.
- The values of delay-related counters or indicators such as **N.QoS.DL.PktDelayAirInterface.Time** may increase in the cell involved.
- The UE traffic distribution across frequency bands changes. As a result, the values of all related band-specific counters change.
- In inter-base-station CA scenarios, the average downlink throughput of CA UEs decreases.

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-070201	Multi-Frequency Spatial Smart Carrier Selection	NR0S00SFBS00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgorithmSwitch parameter	<i>Carrier Aggregation</i>	Intra-base-station inter-carrier frequency selective scheduling (controlled by the CROSS_CARRIER_FREQ_SEL_SCH_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) can be enabled only after intra-band CA is enabled.
Low-frequency TDD	Intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgorithmSwitch parameter	<i>Carrier Aggregation</i>	Intra-base-station inter-carrier frequency selective scheduling (controlled by the CROSS_CARRIER_FREQ_SEL_SCH_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) can be enabled only after intra-FR inter-band CA is enabled.
Low-frequency TDD	Intra-base-station inter-carrier frequency selective scheduling	CROSS_CARRIER_FREQ_SEL_SCH_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	Inter-base-station inter-carrier frequency selective scheduling (controlled by the CROSS_CARR_FREQ_SEL_SCH_ENH_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) can be enabled only when both intra-base-station inter-carrier frequency selective scheduling and downlink fast inter-gNodeB CA (controlled by the INTER_GNB_ULTRALOW_LAT_SPLT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) are enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink fast inter-gNodeB CA	INTER_GNB_ULTRALOW_LAT_SPLT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Carrier Aggregation</i>	Inter-base-station inter-carrier frequency selective scheduling (controlled by the CROSS_CARR_FREQ_SEL_SCH_ENH_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) can be enabled only when both intra-base-station inter-carrier frequency selective scheduling (controlled by the CROSS_CARRIER_FREQ_SEL_SCH_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) and downlink fast inter-gNodeB CA are enabled.

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Energy-efficiency-based coordinated scheduling	ENERGY_EFF_BASED_COLLAB_SCH_SW option of the NRCCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When inter-carrier frequency selective scheduling and energy-efficiency-based coordinated scheduling are both enabled, inter-carrier frequency selective scheduling gains and energy saving gains both decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dynamic RF channel shutdown	RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellAIgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When inter-carrier frequency selective scheduling and dynamic RF channel shutdown are both enabled, inter-carrier frequency selective scheduling gains and energy saving gains both decrease.
Low-frequency TDD	Downlink fast intra-gNodeB CA	CA_ULTRA_LOW_LATENCY_SPLIT_SW option of the NRDUCellCarrierMgmt.EnhancedAlgorithmExtSwitch parameter	<i>CA Experience Improvement</i>	It is recommended that downlink fast intra-gNodeB CA be enabled together with intra-base-station inter-carrier frequency selective scheduling to prevent possible deterioration in the user experience of small-packet UEs.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. The DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Networking

The network uses at least two frequencies.

4.3.5 Others

None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment. The following parameters do not take effect for SUL cells, for which the **NRDUCell.DuplexMode** parameter is set to **CELL_SUL**.

Table 4-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
CA Enhanced Algorithm Extended Switch	NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch	CROSS_CARRIER_FREQ_SEL_SCH_SW	Select this option if intra-base-station inter-carrier frequency selective scheduling is required.
CA Enhanced Algorithm Extended Switch	NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch	AVG_LOAD_BASED_DATA_SPLIT_SW	You are advised to select this option.
CA A5 RSRP Threshold 2	NRCeIlCaMgmtConfig.CaA5RsrpThld2	N/A	If the CROSS_CARRIER_FREQ_SEL_SCH_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter is selected, it is recommended that this parameter be set to a value less than SCells' values of the NRCeIlInterFHoMeaGrp.CovInterFreqA2RsrpThld parameter.

Parameter Name	Parameter ID	Option	Setting Notes
CA A2 RSRP Threshold	NRCellCaMgmtCon- fig.CaA2RsrpThld	N/A	If the CROSS_CARRIER_FREQ_SEL_SCH_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter is selected, it is recommended that this parameter be set to a value less than the value of the NRCellCaMgmtCon- fig.CaA5RsrpThld2 parameter.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

For intra-base-station inter-carrier frequency selective scheduling: The switch controlling intra-base-station inter-carrier frequency selective scheduling must be turned on for the PCell so that this function can take effect. To facilitate future deployment of inter-base-station inter-carrier frequency selective scheduling in contiguous areas, it is recommended that this switch be also turned on for the associated SCells when this function is configured. The following uses intra-base-station 2CC aggregation as an example, with **NrDuCellId** of the PCell being **0** and **NrDuCellId** of the SCell being **1**.

```
//Turning on the switch controlling intra-base-station inter-carrier frequency selective scheduling and the
switch controlling average-load-based data split
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoExtSwitch=CROSS_CARRIER_FREQ_SEL_SCH_SW-1&AVG_LOAD_BASED_DATA_SPLIT_SW-1;
MOD NRDUCELLCARRMGMT: NrDuCellId=1,
CaEnhancedAlgoExtSwitch=CROSS_CARRIER_FREQ_SEL_SCH_SW-1&AVG_LOAD_BASED_DATA_SPLIT_SW-1;
//Setting CA A5 RSRP Threshold 2 and the CA A2 RSRP Threshold
MOD NRCELLCAMGMTCONFIG: NrCellId=0, CaA2RsrpThld=-114, CaA5RsrpThld2=-110;
MOD NRCELLCAMGMTCONFIG: NrCellId=1, CaA2RsrpThld=-114, CaA5RsrpThld2=-110;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Disabling intra-base-station inter-carrier frequency selective scheduling

```
//Turning off the switch controlling intra-base-station inter-carrier frequency selective scheduling for both the PCell and SCell  
MOD NRDUCELLCARRMGMT: NrDuCellId=0,  
CaEnhancedAlgoExtSwitch=CROSS_CARRIER_FREQ_SEL_SCH_SW-0;  
MOD NRDUCELLCARRMGMT: NrDuCellId=1,  
CaEnhancedAlgoExtSwitch=CROSS_CARRIER_FREQ_SEL_SCH_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

If the average downlink UE throughput (indicated by **User Downlink Average Throughput (DU)**) on the entire network increases after inter-carrier frequency selective scheduling is enabled, inter-carrier frequency selective scheduling has taken effect.

4.4.3 Network Monitoring

Observe the average downlink UE throughput to check whether the UE data rate increases on the entire network after this function is enabled.

The average downlink UE throughput is indicated by **User Downlink Average Throughput (DU)**.

5 Smart Inter-Frequency Channel Assistance (Trial)

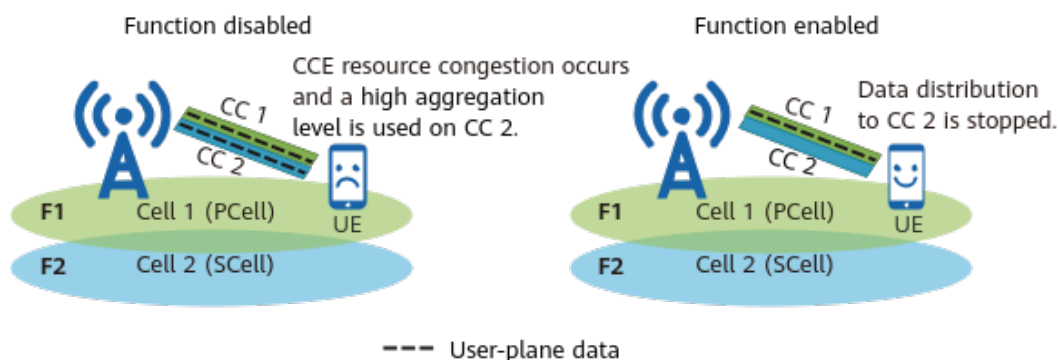
5.1 CCE-based Data Split Stop for CA UEs

5.1.1 Principles

The maximum number of CCEs available in an NR cell depends on the cell bandwidth, subcarrier spacing, and number of PDCCH symbols. Cell-edge UEs use higher CCE aggregation levels than cell-center UEs and consume more CCEs on average. When there is a large number of cell-edge UEs running small-packet services, CCEs are likely to be insufficient. As a result, UE scheduling may fail, affecting network performance.

To address this issue, CCE-based data split stop for CA UEs is introduced. As illustrated in [Figure 5-1](#), if CCE resource congestion occurs in an NR cell, the gNodeB selects some CA UEs that use high CCE aggregation levels and stops data distribution to the carrier in the congestion state for the selected UEs. This allows some CCEs to be released, thereby increasing the average cell throughput. This function is controlled by the **CCE_BASED_CA_SPLIT_STOP_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter. CCE-based data split stop for CA UEs is a trial function.

Figure 5-1 Schematic diagram of CCE-based data split stop for CA UEs



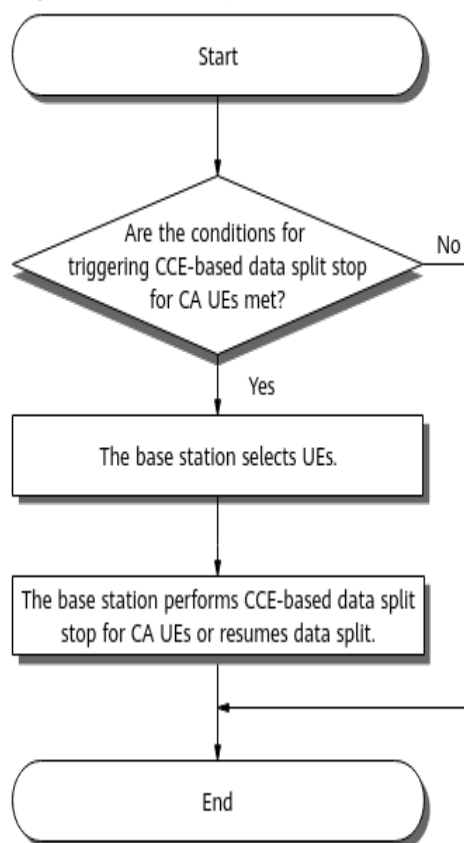
NOTE

Trial functions are functions that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these functions can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial functions shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial functions. Trial functions are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial functions may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial functions but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial functions are free, Huawei is not liable for any trial function malfunctions or any losses incurred by using the trial functions. Huawei does not promise that problems with trial functions will be resolved in the current version. Huawei reserves the rights to convert trial functions into commercial functions in later R/C versions. If trial functions are converted into commercial functions in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial functions. If a customer fails to purchase such a license, the trial function(s) will be invalidated automatically when the product is upgraded.

Figure 5-2 shows the basic process of CCE-based data split stop for CA UEs.

Figure 5-2 Basic process of CCE-based data split stop for CA UEs



1. The scenarios where CCE-based data split stop for CA UEs can be triggered are described in [5.1.1.1 Triggering of CCE-based Data Split Stop for CA UEs](#).
2. How the gNodeB selects UEs for data split stop is described in [5.1.1.2 UE Selection](#).

3. How the gNodeB determines whether to stop or resume data split is described in [5.1.1.3 CCE-based Data Split Stop and Resumption for CA UEs](#).

5.1.1.1 Triggering of CCE-based Data Split Stop for CA UEs

Each NR cell checks its CCE congestion status every second, based on which the gNodeB determines whether to trigger CCE-based data split stop for CA UEs. This function can be triggered when all the following conditions are met:

In the case of an NR TDD cell:

- The downlink CCE allocation failure rate, which is the result of $(N.CCE.DL.AllocReq.Num - N.CCE.DL.AggLvl1Num - N.CCE.DL.AggLvl2Num - N.CCE.DL.AggLvl4Num - N.CCE.DL.AggLvl8Num - N.CCE.DL.AggLvl16Num) / N.CCE.DL.AllocReq.Num$, is greater than or equal to 30%.
- If PDCCH symbol number adaptation (controlled by the **UE_PDCCH_SYM_NUM_ADAPT_SW** option of the **NRDUCellPdcch.PdcchAlgoExtSwitch** parameter) is enabled, the number of PDCCH symbols must reach the maximum allowed fixed value. If PDCCH symbol number adaptation is disabled, the number of PDCCH symbols does not need to be checked. As PDCCH symbol number adaptation increases the amount of available CCE resources, it is recommended that PDCCH symbol number adaptation be enabled before CCE-based data split stop for CA UEs is enabled. For details about PDCCH symbol number adaptation, see *Channel Management*.
- If uplink DCI extension-based scheduling is enabled (controlled the **DCI_EXTEND_UL_CONGEST_SW** option of the **NRDUCellPusch.UlPuschAlgoSwitch** parameter), uplink DCI extension-based scheduling needs to take effect. If uplink DCI extension-based scheduling is disabled, whether uplink DCI extension-based scheduling takes effect does not need to be checked. As uplink DCI extension-based scheduling increases the amount of available CCE resources, it is recommended that uplink DCI extension-based scheduling be enabled before CCE-based data split stop for CA UEs is enabled. For details about uplink DCI extension-based scheduling, see *Uplink Scheduling*.

NOTE

It is recommended that the initial CCE aggregation level selection optimization function (controlled by the **AGG_LVL_INIT_SELECT_OPT_SW** option of the **NRDUCellPdcch.PdcchAlgoEnhSwitch** parameter) be enabled and take effect normally before CCE-based data split stop for CA UEs is enabled. Otherwise, the CCE aggregation levels for UEs may be high, causing PDCCH resource congestion. For details about the initial CCE aggregation level selection optimization function, see *Channel Management*.

5.1.1.2 UE Selection

After CCE-based data split stop for CA UEs is triggered, UE selection can be triggered only when the following condition is met:

The downlink PRB usage of the carrier in the congestion state is less than or equal to 70%. The downlink PRB usage is equal to the result of **N.PRB.DL.Used.Avg** divided by **N.PRB.DL.Avail.Avg**.

When the preceding condition is met, this function can take effect for any UE that meets all of the following conditions:

- The UE is a CA UE for which data is being distributed.
- CQIs reported by the UE have been received by the gNodeB.
- The CCE aggregation level for the UE is greater than or equal to 8.
- No other data split stop functions are taking effect for the UE.

5.1.1.3 CCE-based Data Split Stop and Resumption for CA UEs

Data Split Stop

After UEs are selected for the carrier where data split stop is triggered, the gNodeB notifies the PCC of the selection result. Then, the PCC determines whether to stop data split based on the following conditions. If all the following conditions are met, the gNodeB stops distributing data to the carrier in the congestion state, which can be the PCC or an SCC.

- The proportion of data to be distributed to the carrier where data split stop is triggered is less than or equal to 30%.
- The downlink PRB usage of at least one of the carriers where data split stop is not triggered is less than or equal to 60%.
- In the case of an NR TDD cell, the downlink CCE allocation failure rate of at least one of the carriers where data split stop is not triggered is less than 10%.

NOTE

The proportion of data to be distributed to a carrier for a UE is determined by the ratio of the comprehensive scheduling capability of the carrier to the sum of the comprehensive scheduling capabilities of all carriers for the UE. The proportion is maintained by the PCC. The calculation of the comprehensive scheduling capability of the carrier is related to the load of the carrier. The load statistics vary depending on the setting of the **AVG_LOAD_BASED_DATA_SPLIT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter.

- If this option is deselected, the average load of the PCell and SCells within a measurement period will not be considered.
- If this option is selected, the average load of the PCell and SCells within a measurement period will be considered. This statistics method is superior. Therefore, it is recommended that this option be selected when CCE-based data split stop for CA UEs is enabled.

Data Split Resumption

When the condition for cell-level or UE-level data split resumption is met, data split will be resumed.

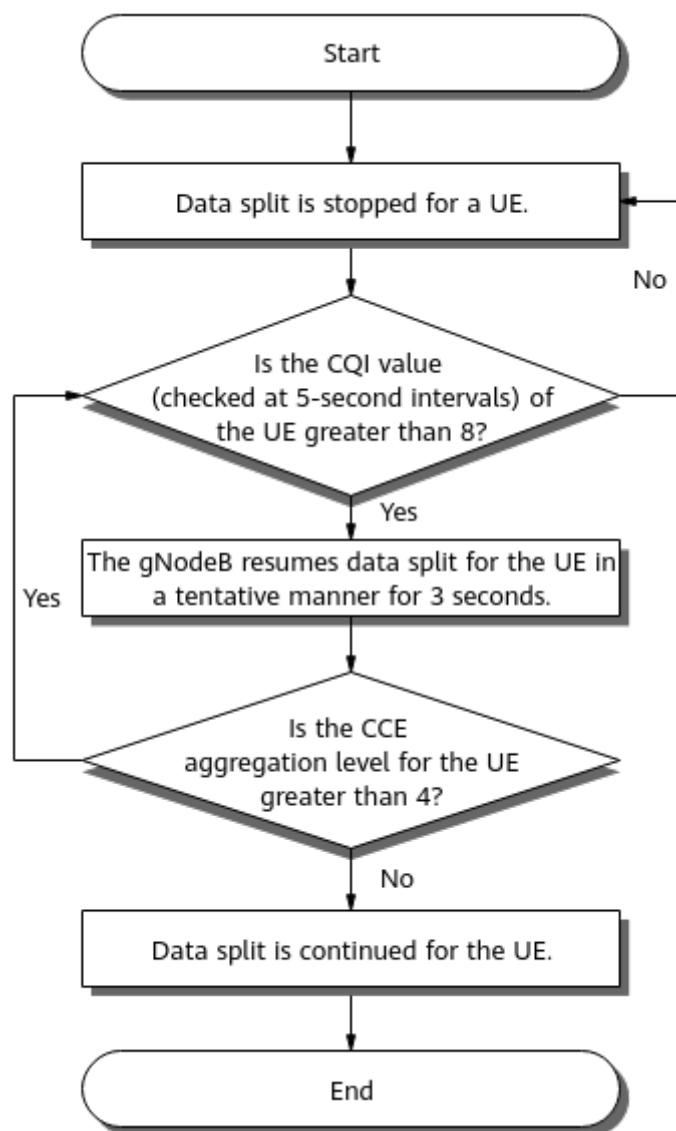
- Cell-level data split resumption:

The NR cell periodically checks its CCE congestion status, based on which the gNodeB determines whether to exit CCE-based data split stop for CA UEs.

In the case of an NR TDD cell, if the downlink CCE allocation failure rate of the carrier to which data stops being distributed is less than 10%, the cell exits the CCE congestion state, and data split will be gradually resumed for the UEs involved.

- UE-level data split resumption:
Figure 5-3 shows the process in which the gNodeB determines whether to resume data split for a UE.

Figure 5-3 Process of UE-level data split resumption



Details of UE-level data split resumption are as follows:

After data split is stopped for a UE, the gNodeB checks the CQI value of the UE at 5-second intervals.

- If the CQI value is greater than 8, the gNodeB resumes data split for the UE in a tentative manner for 3 seconds and then makes a further decision as follows:
 - If the CCE aggregation level for the UE is less than or equal to 4, data split is continued.
 - If the CCE aggregation level for the UE is greater than 4, another tentative data split resumption attempt is made.

- If the CQI value is less than or equal to 8, the gNodeB continues to stop data split for the UE and waits for the next tentative data split resumption period.

5.1.2 Network Analysis

5.1.2.1 Benefits

When downlink or uplink CCE resources are congested, CCE-based data split stop for CA UEs increases the **Cell Downlink Average Throughput (DU)**, **Cell Uplink Average Throughput (DU)**, **User Downlink Average Throughput (DU)**, and average uplink CEU throughput, which is the result of $(N.ThpVol.UL.Edge - N.ThpVol.UE.UL.SmallPkt.Edge) / N.ThpTime.UE.UL.RmvSmallPkt.Edge$. The higher the proportion of cell-edge small-packet CA UEs with low scheduling priorities, the greater the benefits.

If the proportion of small-packet UEs or cell-edge UEs is excessively high in an NR cell in the CCE resource congestion state, there are no significant benefits for the average cell throughput, and the average UE throughput remains unchanged or slightly decreases. In such scenarios, CCE-based data split stop for CA UEs is not recommended.

5.1.2.2 Impacts

The impacts between this function and other functions are described in [5.1.3.2.3 Function Impacts](#).

CCE-based data split stop for CA UEs may have the following impacts on the network:

- The downlink PRB usage of the cell where data split is stopped (indicated by **N.PRB.DL.Used.Avg** divided by **N.PRB.DL.Avail.Avg**) increases.
- The average number of UEs in RRC_CONNECTED mode in the cell where data split is stopped (indicated by **N.User.RRCCConn.Avg**) increases.
- If data split is stopped on the PCC, the downlink MAC-layer traffic volume of CA UEs that treat the local cell as their PCell (indicated by **N.CA.PCell.ThpVol.DL.Cell**) decreases.
- If data split is stopped on the SCC, the downlink MAC-layer traffic volume of CA UEs that treat the local cell as their SCell (indicated by **N.CA.SCell.ThpVol.DL.Cell**) decreases.
- The average downlink throughput of a CA UE selected for data split stop slightly decreases if CCE resource congestion occurs on the carriers where data split is not stopped and the cells using these carriers are in the medium or heavy load state. The decrease can be up to 30%.

5.1.3 Requirements

5.1.3.1 Licenses

There are no license requirements for trial functions.

5.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	CCE-based data split stop for CA UEs (controlled by the CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) requires at least one of the downlink intra-band CA and downlink intra-FR inter-band CA functions to be enabled.
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	

5.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	High-speed cells (for which NRDUCell.HighSpeedFlag is set to HIGH_SPEED) do not support CCE-based data split stop for CA UEs (controlled by the CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrMgmt.CaEnhancedAgoExtSwitch parameter).
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Hyper cells do not support CCE-based data split stop for CA UEs (controlled by the CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrMgmt.CaEnhancedAgoExtSwitch parameter).
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE NRDUCell.MultiTrp.DataTranMode set to BASIC_MODE or DPS_MODE	<i>Cell Combination</i>	Combined cells do not support CCE-based data split stop for CA UEs (controlled by the CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrMgmt.CaEnhancedAgoExtSwitch parameter).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Joint Transmission Cell Combination	- NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE - NRDUCellMultiTrp.DataTranMode set to JOINT_MODE	Cell Combination	Combined cells in joint transmission mode do not support CCE-based data split stop for CA UEs (controlled by the CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter).

5.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	Carrier Aggregation	CCE-based data split stop for CA UEs (controlled by the CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) takes effect only for UEs in the downlink CA state. After this function takes effect, data scheduling for CA UEs with SCells activated may accelerate and traffic volume-triggered SCell deactivation may be more likely to take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink intra-band CA	INTRA_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	CCE-based data split stop for CA UEs (controlled by the CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) does not take effect for UEs in the uplink CA state.
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	CCE-based data split stop for CA UEs (controlled by the CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) takes effect only for UEs in the downlink CA state. After this function takes effect, data scheduling for CA UEs with SCells activated may accelerate and traffic volume-triggered SCell deactivation may be more likely to take effect.
Low-frequency TDD	Uplink intra-FR inter-band CA	INTRA_FR_INTER_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	CCE-based data split stop for CA UEs (controlled by the CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) does not take effect for UEs in the uplink CA state.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Initial CCE aggregation level selection optimization	AGG_LVL_INIT_SELECT_OPT_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>Channel Management</i>	When CCE-based data split stop for CA UEs (controlled by the CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) is enabled, it is recommended that initial CCE aggregation level selection optimization (controlled by the AGG_LVL_INIT_SELECT_OPT_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter) be also enabled. This optimizes PDCCH resource allocation by lowering the CCE aggregation level when the cell load is heavy.
Low-frequency TDD	Downlink fast inter-gNodeB CA	INTER_GNB_ULTRALOW_LAT_SPLT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>CA Experience Improvement</i>	When CCE-based data split stop for CA UEs (controlled by the CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) is enabled, it is recommended that downlink fast inter-gNodeB CA be also enabled to improve the flexibility of stopping data distribution to inter-gNodeB SCells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multi-frequency convergent scheduling	MULTI_FREQ_CONVERGENT_SCH_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	When multi-frequency convergent scheduling (controlled by the MULTI_FREQ_CONVERGENT_SCH_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch parameter) and CCE-based data split stop for CA UEs are enabled together, CCE-based data split stop for CA UEs preferentially takes effect in downlink data split for CA UEs. This compromises the benefits of multi-frequency convergent scheduling.
Low-frequency TDD	Single DCI	DL_SINGLE_DCI_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	Single DCI takes precedence over CCE-based data split stop for CA UEs. If single DCI has taken effect for a UE, CCE-based data split stop for CA UEs will not take effect for this UE. If single DCI is triggered for a UE for which CCE-based data split stop for CA UEs has taken effect, CCE-based data split stop for CA UEs will be terminated for this UE.

5.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.1.3.4 Networking

- NR TDD frequencies are deployed on the network.
- The Xn interface between gNodeBs works properly.

5.1.4 Operation and Maintenance

5.1.4.1 Data Configuration

5.1.4.1.1 Data Preparation

Table 5-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
CA Enhanced Algorithm Extended Switch	NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch	CCE_BASED_CA_SPLIT_STOP_SW	You are advised to select this option.

Parameter Name	Parameter ID	Option	Setting Notes
CA Enhanced Algorithm Extended Switch	NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch	AVG_LOAD_BASE D_DATA_SPLIT_SW	You are advised to select this option.
PDCCH Algorithm Extension Switch	NRDUCellPdcch.PdcchAlgoExtSwitch	UE_PDCCH_SYM_NUM_ADAPT_SW	You are advised to select this option.
Uplink PUSCH Algorithm Switch	NRDUCellPusch.UlPuschAlgoSwitch	DCI_EXTEND_UL_CONGEST_SW	You are advised to select this option.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcch.PdcchAlgoEnhSwitch	AGG_LVL_INIT_SELECT_OPT_SW	You are advised to select this option.
CA Enhanced Algorithm Extended Switch	NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch	INTER_GNB_ULTRALLOW_LAT_SPLT_SW	You are advised to select this option.
CA Algorithm Switch	NRDUCellAlgoSwitch.CaAlgoSwitch	INTRA_BAND_CA_SW	You are advised to select this option.
CA Algorithm Switch	NRDUCellAlgoSwitch.CaAlgoSwitch	INTRA_FR_INTER_BAND_CA_SW	You are advised to select this option.

5.1.4.1.2 Using MML Commands

Before using MML commands, refer to [5.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink intra-band CA for both the PCell and the SCell (using inter-base-station 2CC aggregation as an example)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CaAlgoSwitch=INTRA_BAND_CA_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, CaAlgoSwitch=INTRA_BAND_CA_SW-1;
//Enabling downlink intra-FR inter-band CA for both the PCell and the SCell (using inter-base-station 2CC aggregation as an example)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CaAlgoSwitch=INTRA_FR_INTER_BAND_CA_SW-1;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, CaAlgoSwitch=INTRA_FR_INTER_BAND_CA_SW-1;
//Enabling downlink fast inter-gNodeB CA for both the PCell and the SCell (using inter-base-station 2CC aggregation as an example) (It is recommended that downlink fast inter-gNodeB CA be enabled to improve the flexibility of stopping data distribution to inter-gNodeB SCells.)
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoExtSwitch=INTER_GNB_ULTRALOW_LAT_SPLT_SW-1;
MOD NRDUCELLCARRMGMT: NrDuCellId=1,
CaEnhancedAlgoExtSwitch=INTER_GNB_ULTRALOW_LAT_SPLT_SW-1;
// (Recommended when the cell is heavily loaded) Enabling PDCCH symbol number adaptation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoExtSwitch=UE_PDCCH_SYM_NUM_ADAPT_SW-1;
MOD NRDUCELLPDCCH: NrDuCellId=1, PdcchAlgoExtSwitch=UE_PDCCH_SYM_NUM_ADAPT_SW-1;
// (Recommended when the cell is heavily loaded) Enabling uplink DCI extension-based scheduling
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPuschAlgoSwitch=DCI_EXTEND_UL_CONGEST_SW-1;
MOD NRDUCELLPUSCH: NrDuCellId=1, UIPuschAlgoSwitch=DCI_EXTEND_UL_CONGEST_SW-1;
// (Recommended when the cell is heavily loaded) Enabling initial CCE aggregation level selection optimization
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=AGG_LVL_INIT_SELECT_OPT_SW-1;
MOD NRDUCELLPDCCH: NrDuCellId=1, PdcchAlgoEnhSwitch=AGG_LVL_INIT_SELECT_OPT_SW-1;
//Selecting the CCE_BASED_CA_SPLIT_STOP_SW and AVG_LOAD_BASED_DATA_SPLIT_SW options
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoExtSwitch=CCE_BASED_CA_SPLIT_STOP_SW-1&AVG_LOAD_BASED_DATA_SPLIT_SW-1;
MOD NRDUCELLCARRMGMT: NrDuCellId=1,
CaEnhancedAlgoExtSwitch=CCE_BASED_CA_SPLIT_STOP_SW-1&AVG_LOAD_BASED_DATA_SPLIT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling CCE-based data split stop for CA UEs
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoExtSwitch=CCE_BASED_CA_SPLIT_STOP_SW-0;
MOD NRDUCELLCARRMGMT: NrDuCellId=1, CaEnhancedAlgoExtSwitch=CCE_BASED_CA_SPLIT_STOP_SW-0;
```

5.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.1.4.2 Activation Verification

After CCE-based data split stop for CA UEs is enabled, if the downlink MAC-layer traffic volume of CA UEs in the cell where data split is stopped decreases, this function has taken effect.

- The downlink MAC-layer traffic volume of CA UEs in the PCell where data split is stopped is measured by **N.CA.PCell.ThpVol.Dl.Cell**.
- The downlink MAC-layer traffic volume of CA UEs in an SCell where data split is stopped is measured by **N.CA.SCell.ThpVol.Dl.Cell**.

5.1.4.3 Network Monitoring

Observe **Cell Downlink Average Throughput (DU)** to monitor the increase in the downlink cell throughput.

Observe **Cell Uplink Average Throughput (DU)** to monitor the increase in the uplink cell throughput.

Observe **User Downlink Average Throughput (DU)** to monitor the increase in the downlink UE throughput.

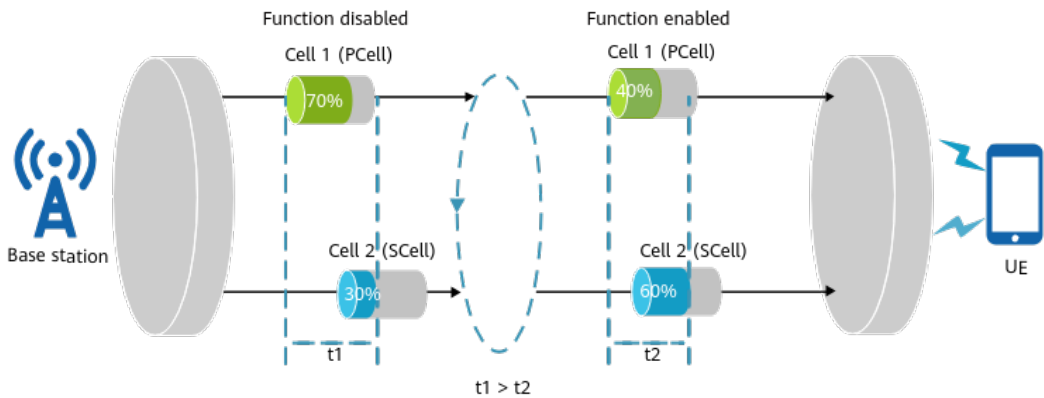
Check the result of $(N.ThpVol.UL.Edge - N.ThpVol.UE.UL.SmallPkt.Edge) / N.ThpTime.UE.UL.RmvSmallPkt.Edge$ to monitor the increase in the uplink cell-edge UE throughput.

6 Multi-Frequency Joint Scheduling

6.1 Principles

In multi-carrier scenarios, the downlink scheduling of a CA UE in its PCell and SCells is relatively independent of each other. As a result, RB resources cannot precisely match traffic volumes. In addition, the traditional downlink scheduling solution considers only one of the time and frequency domains. To address this issue, the multi-frequency joint scheduling function is introduced. This function achieves precise scheduling for CA UEs and optimizes downlink scheduling in both the time and frequency domains, increasing the average downlink throughput of UEs. How this function works is illustrated in [Figure 6-1](#).

Figure 6-1 Schematic diagram of multi-frequency joint scheduling



Multi-frequency joint scheduling includes the functions described in [Table 6-1](#), and it is recommended that all the functions be enabled.

Table 6-1 Functions of multi-frequency joint scheduling

Function Name	Function Switch	Working Principles
Multi-frequency convergent scheduling	MULTI_FREQ_CONVERGENT_SWITCH_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoExtSwitch parameter	Multi-frequency convergent scheduling is supported only in intra-base-station CA scenarios. It supports real-time CA UE scheduling, in which the PCell and SCells can obtain the buffered traffic volume in real time and the PCell can allocate PUCCH resources to the SCells in real time. This allows the amount of scheduled data to accurately match the scheduling capability. This function is supported only in intra-BBP CA scenarios.
CA concurrent scheduling capability increase	CA_CONCURRENT_SCH_CAPB_INCR_SW option of the gNodeBAlgo.DLSchSwitch parameter	The CA concurrent scheduling capability increase function enables more CA-SCell-activated UEs to be concurrently scheduled.
CA Performance Boost Based on Multi-dimensional Optimization	CA_PRFM_BOOST_BSD_ON_MD_OPT_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoExtSwitch parameter	The CA Performance Boost Based on Multi-dimensional Optimization function is supported only in intra-base-station CA scenarios on TDD networks. The gNodeB dynamically evaluates the scheduling capability of each carrier in all its served cells with CA Performance Boost Based on Multi-dimensional Optimization enabled. The factors considered in the evaluation include the service data volumes and spectral efficiency of CA UEs. Based on the evaluation result, the gNodeB adjusts the carrier-specific data split ratio and scheduling priority for CA UEs to optimize the amount of service data to be distributed and improve the UE scheduling efficiency, thereby increasing the average downlink throughput of CA UEs. This function is supported in both intra-BBP CA and inter-BBP CA scenarios.

 NOTE

Intra-BBP CA refers to when the PCell and SCells of a CA UE are deployed on the same BBP, and inter-BBP CA refers to when the PCell and SCells of a CA UE are deployed on different BBPs.

6.2 Network Analysis

6.2.1 Benefits

When all the sub-functions of the multi-frequency joint scheduling function take effect, the average downlink throughput of UEs increases.

It is recommended that the CA Performance Boost Based on Multi-dimensional Optimization function be enabled for all cells in an area. Otherwise, the maximum gains in the average downlink throughput of UEs may not be obtained.

The benefits are related to the following factors:

- The larger the number of carriers on the network and the larger the bandwidth of each carrier, the greater the benefits.
- When the network load is neither light nor heavy, the higher the network load, the greater the benefits.
- The greater the load difference between frequency bands, the greater the benefits.
- The greater the bandwidth difference between frequency bands, the greater the benefits. If the bandwidths are the same, the benefits may decrease.
- When multiple frequency bands are deployed on the same BBP, the benefits are greater.
- The CA UE traffic proportion is more than 20% of the total traffic on the network.

CA UE traffic proportion = $(N.CA.SccActiveUser.ThpVol.DL / N.ThpVol.DL) \times 100\%$

6.2.2 Impacts

The impacts between this function and other functions are described in [6.3.2.3 Function Impacts](#).

When all the sub-functions of the multi-frequency joint scheduling function take effect, there may be the following impacts on the network:

- The downlink PRB usage of the cell involved (indicated by the result of **N.PR.B.DL.Used.Avg** divided by **N.PR.B.DL.Avail.Avg**) increases.
- The sizes of the last packets in the PCell and SCells (which can be observed using the **N.CA.ThpVol.DL.LastSlot** counter) change. When the sizes of the last packets decrease, the MCS indexes for small packets will be more frequently decreased. As a result, the average MCS index in the PCell and SCells decreases, and the average downlink throughput in the PCell and SCells (indicated by the result of **N.ThpVol.DL.Cell** divided by **N.ThpTime.DL.Cell**) decreases.
- If the spectral efficiency of the SCells is higher than that of the PCell before this feature is enabled, the volume of traffic scheduled in the PCell decreases and the volume of traffic scheduled in the SCells increases after this feature is enabled. As a result, the average downlink throughput (indicated by the result

of **N.ThpVol.DL.Cell** divided by **N.ThpTime.DL.Cell**) of the PCell decreases and that of the SCells increases.

- After this feature is enabled, the CA- and non-CA-related traffic distributions in the cell involved change. As a result, the values of CA-related counters or indicators also change. For example, the values of **N.CA.PR.B.DL.PCell.Used.Avg** and **N.CA.PR.B.DL.SCell.Used.Avg**, and the average throughput of CA UEs with SCells activated will change. The average throughput of CA UEs with SCells activated is the result of $(\text{N.CA.SccActiveUser.ThpVol.DL} - \text{N.CA.ThpVol.DL.LastSlot}) / \text{N.CA.ThpTime.DL.RmvLastSlot}$.
- After this feature is activated, the average downlink throughput of CA UEs increases and the amount of data in the downlink RLC buffer for these UEs decreases in intra-BBP CA scenarios. As a result, the numbers of SCell activation executions, successful SCell activation executions, SCell deactivation executions, and successful SCell deactivation executions for CA UEs change. The changes in turn cause the values of the counters related to CA UEs with activated SCells to change, such as **N.User.CA.PCell.DL.SCellAct.Avg** (which measures the average number of downlink CA UEs that treat the local cell as their PCell and have their activated SCells) and the counters related to PCell and SCell MAC measurements for CA UEs with activated SCells. For details about the counters related to CA UEs with activated SCells, see *Carrier Aggregation*.
- The UE traffic distribution across frequency bands changes. As a result, the values of all related band-specific counters change.
- In inter-base-station CA scenarios, the average downlink throughput of CA UEs decreases.
- After CA Performance Boost Based on Multi-dimensional Optimization takes effect, the average transmission delay of the first packets (result of $\text{N.Traffic.DL.RlcFirstPktDelay.Time} / \text{N.Qos.DL.RlcFirstPktDelay.Num}$) increases.

6.3 Requirements

6.3.1 Licenses

The following license control item is required for the multi-frequency convergent scheduling and CA Performance Boost Based on Multi-dimensional Optimization functions in TDD.

Feature ID	Feature Name	Model	Sales Unit
FOFD-081205	CA Performance Boost Based on Multi-dimensional Optimization	NR0S00MCEA00	per Cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

- Multi-frequency convergent scheduling

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-band CA	INTRA_BAND_CA_SWITCH option of the NRDUCellAlgoSwitch parameter	<i>Carrier Aggregation</i>	Multi-frequency convergent scheduling can take effect only when intra-band CA or intra-FR inter-band CA is enabled.
Low-frequency TDD	Intra-FR inter-band CA	INTRA_FRIINTER_BAND_CA_SWITCH option of the NRDUCellAlgoSwitch parameter	<i>Carrier Aggregation</i>	

- CA concurrent scheduling capability increase

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	CA concurrent scheduling capability increase can take effect only when intra-band CA or intra-FR inter-band CA is enabled.
Low-frequency TDD	Intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	

- CA Performance Boost Based on Multi-dimensional Optimization

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	CA Performance Boost Based on Multi-dimensional Optimization can be enabled only when intra-band CA or intra-FR inter-band CA has been enabled.
Low-frequency TDD	Intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	

6.3.2.2 Mutually Exclusive Functions

- Multi-frequency convergent scheduling

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	- NRDUCell.NrDualCellNetworkingMode set to HYPER_CELL_COMBINE_MODE - NRDUCell.MultiTrp.DataTransMode set to BASIC_MODE or DPS_MODE	Cell Combination	Combined cells do not support multi-frequency convergent scheduling.

- CA concurrent scheduling capability increase
None
- CA Performance Boost Based on Multi-dimensional Optimization

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	- NRDUCell.NrDualCellNetworkingMode set to HYPER_CELL_COMBINE_MODE - NRDUCell.MultiTrp.DataTransMode set to BASIC_MODE or DPS_MODE	Cell Combination	Combined cells do not support CA Performance Boost Based on Multi-dimensional Optimization.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Joint Transmission Cell Combination	- NRDUCell.NrDuCellNetworkWorkingMode set to HYPER_CELL_COMBINE_MODE - NRDUCellMultiTrp.DataTransMode set to JOINT_MODE	Cell Combination	Combined cells in joint transmission mode do not support CA Performance Boost Based on Multi-dimensional Optimization.
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkWorkingMode set to HYPER_CELL	Hyper Cell	Hyper cells do not support CA Performance Boost Based on Multi-dimensional Optimization.
Low-frequency TDD	Fusion Cell	NRDUCell.NrDuCellNetworkWorkingMode set to FUSION_CELL	Fusion Cell (Low-Frequency TDD)	If Fusion Cell has been enabled, CA Performance Boost Based on Multi-dimensional Optimization cannot be enabled.
Low-frequency TDD	Energy-efficiency-based coordinated scheduling	ENERGY_EFF_BASED_COLLAB_SWITCH option of the NRCCellAlgoSwitch.MultiFreqAlgoSwitch parameter	Energy Conservation and Emission Reduction	If energy-efficiency-based coordinated scheduling has been enabled, CA Performance Boost Based on Multi-dimensional Optimization cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station multi-user Co-MM	INTRA_GNB_SU_COM_M_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	None	If intra-base-station multi-user Co-MM has been enabled, CA Performance Boost Based on Multi-dimensional Optimization cannot be enabled.

6.3.2.3 Function Impacts

- Multi-frequency convergent scheduling
 - Impact relationships with coordination-related functions are as follows.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COM_M_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	Multi-frequency convergent scheduling delivers fewer benefits when it is enabled together with Co-MM.
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	Multi-frequency convergent scheduling delivers fewer benefits when it is enabled together with intra-base-station DL CoMP.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	- Multi-frequency convergent scheduling enables real-time scheduling for individual CA UEs to resolve the problem of inaccurate downlink data split between the PCell and SCells. - NR multi-band power adaptive scheduling allocates the estimated remaining power of a carrier to other carriers for a CA UE if the carrier is not allocated enough data as a result of inaccurate downlink data split between the PCell and SCells. Therefore, both multi-frequency convergent scheduling and NR multi-band power adaptive scheduling can improve the downlink experience of

RAT	Function Name	Function Switch	Reference	Description
				CA UEs. When both functions are enabled, the gains provided by either function may decrease.

- Impact relationships with functions related to carrier management are as follows.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-gNodeB CA	INTER_GNODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>CA Experience Improvement</i>	The more UEs in the inter-gNodeB CA state, the smaller the gains delivered by multi-frequency convergent scheduling.
Low-frequency TDD	CCE-based data split stop for CA UEs	CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	When both multi-frequency convergent scheduling and CCE-based data split stop for CA UEs are enabled, CCE-based data split stop for CA UEs preferentially takes effect in downlink data split for CA UEs, compromising the benefits of multi-frequency convergent scheduling.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink fast intra-gNodeB CA	CA_ULTRA_LOW_LATE_NCY_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>CA Experience Improvement</i>	- Multi-frequency convergent scheduling not only enables data transmission to start simultaneously in SCells and the PCell, but also makes CA scheduling more precise. Therefore, if downlink fast intra-gNodeB CA is enabled after multi-frequency convergent scheduling takes effect, downlink fast intra-gNodeB CA does not offer any benefits. - If multi-frequency convergent scheduling is enabled after downlink fast intra-gNodeB CA takes effect, multi-frequency convergent scheduling does not offer the benefits

RAT	Function Name	Function Switch	Reference	Description
				brought by simultaneous data transmissions in the PCell and SCells but offers only the benefits brought by improved CA UE scheduling accuracy.

- Impact relationships with other functions are as follows.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Energy-efficiency-based coordinated scheduling	ENERGY_EFF_BASED_COLLAB_SCH_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Multi-frequency convergent scheduling delivers fewer benefits when it is enabled together with energy-efficiency-based coordinated scheduling.
Low-frequency TDD	PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	When multi-frequency convergent scheduling and PDSCH MU-MIMO are both enabled and a high proportion of UEs are paired for PDSCH MU-MIMO, the gains of multi-frequency convergent scheduling decrease.

- CA concurrent scheduling capability increase
None
- CA Performance Boost Based on Multi-dimensional Optimization
 - Impact relationships with functions related to power saving are as follows.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	CA Performance Boost Based on Multi-dimensional Optimization does not take effect for cells in the shutdown state.
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTICARRIER_SD_SWITCH or INTER_GNB_MULTICARRIER_SD_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	CA Performance Boost Based on Multi-dimensional Optimization does not take effect for cells in the shutdown state.
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCYSW set to a value other than OFF	<i>Multi-RAT Coordinated Energy Saving</i>	CA Performance Boost Based on Multi-dimensional Optimization does not take effect for cells in the shutdown state.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR intelligent carrier shutdown	LNR_SMAR T_CARRIER _SHUTDO WN_SW option of the NRDUCell AlgoSwitc h.PowerSa vingSwitch parameter	<i>Multi-RAT Coordinate d Energy Saving</i>	CA Performance Boost Based on Multi-dimensional Optimization does not take effect for cells in the shutdown state.
Low-frequency TDD	pRRU deep dormancy in scheduled mode	RRU.DOR MANCYSW	<i>Multi-RAT Coordinate d Energy Saving</i>	CA Performance Boost Based on Multi-dimensional Optimization does not take effect for cells in the deep dormancy state.
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP _DORM_S W option of the RRUEXTIN FO.RRUDO ROPTFEAT SW parameter	<i>Multi-RAT Coordinate d Energy Saving</i>	RF module fast deep dormancy takes precedence over CA Performance Boost Based on Multi-dimensional Optimization when both functions are enabled. CA Performance Boost Based on Multi-dimensional Optimization does not take effect after RF module fast deep dormancy takes effect.

- Impact relationships with functions related to carrier management are as follows.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-gNodeB CA	INTER_GN ODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>CA Experience Improvement</i>	The more UEs in the inter-gNodeB CA state, the smaller the gains delivered by CA Performance Boost Based on Multi-dimensional Optimization.
Low-frequency TDD	Downlink fast intra-gNodeB CA	CA_ULTRA_LOW_LATE_NCY_SPLIT_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoExtSwitch parameter	<i>CA Experience Improvement</i>	It is recommended that downlink fast intra-gNodeB CA be enabled before CA Performance Boost Based on Multi-dimensional Optimization is enabled. If downlink fast intra-gNodeB CA is not enabled first, the gains of CA Performance Boost Based on Multi-dimensional Optimization may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA concurrent scheduling capability increase	CA_CONCURRENT_SCHEDULING_CAPABILITY_INCREASE_SWITCH option of the gNodeBAdvancedParameters parameter	<i>Multi-Frequency Smart Aggregation</i>	It is recommended that CA concurrent scheduling capability increase be enabled before CA Performance Boost Based on Multi-dimensional Optimization is enabled. If CA concurrent scheduling capability increase is not enabled first, the gains of CA Performance Boost Based on Multi-dimensional Optimization may decrease.

- Impact relationships with other functions are as follows.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COM_M_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	• CA Performance Boost Based on Multi-dimensional Optimization does not take effect for UEs for which Co-MM has taken effect. • On a given gNodeB, if CA Performance Boost Based on Multi-dimensional Optimization is enabled only for some of the cells for which intra-base-station Co-MM is enabled, the number of cells in which CA Performance Boost Based on Multi-dimensional Optimization takes effect may be less than the number of cells for which this function is enabled. Therefore, it is recommended that CA Performance

RAT	Function Name	Function Switch	Reference	Description
				Boost Based on Multi-dimensional Optimization be enabled for all cells under the gNodeB for which intra-base-station Co-MM is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_Switch option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	• CA Performance Boost Based on Multi-dimensional Optimization does not take effect for UEs for which intra-base-station DL CoMP has taken effect. • On a given gNodeB, if CA Performance Boost Based on Multi-dimensional Optimization is enabled only for some of the cells for which intra-base-station DL CoMP is enabled, the number of cells in which CA Performance Boost Based on Multi-dimensional Optimization takes effect may be less than the number of cells for which this function is enabled. Therefore, it is recommended

RAT	Function Name	Function Switch	Reference	Description
				d that CA Performance Boost Based on Multi-dimensional Optimization be enabled for all cells under the gNodeB for which intra-base-station DL CoMP is enabled.
Low-frequency TDD	Percentage-based dynamic RB sharing among operators	RB_DYNAMIC_SHARING_SWITCH option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	<i>Multi-Operator Sharing</i>	If percentage-based dynamic RB sharing among operators is enabled, the scheduling priorities of UEs are preferentially modified based on the policies among operators. In this case, CA Performance Boost Based on Multi-dimensional Optimization provides lower gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Resource-based adaptive scheduling for massive MIMO	NRDUCellPdsch.DLSchOptTimeThld set to a value other than 0	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	If resource-based adaptive scheduling for massive MIMO has been enabled for the cells served by the gNodeB and CA Performance Boost Based on Multi-dimensional Optimization is then also enabled for these cells, the gains of CA Performance Boost Based on Multi-dimensional Optimization may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station CA downlink data split for Cloud X services	CLOUD_X_INTRA_GNB_CA_ENH_SW option of the NRDUCellAlgoSwitch.CloudVrSwitchSwitch parameter	<i>Ultimate Cloud X Experience</i>	While CA Performance Boost Based on Multi-dimensional Optimization aims to improve UE scheduling efficiency and thereby increase the average downlink throughput of CA UEs, intra-base-station CA downlink data split for Cloud X services aims to ensure the delay and user experience of Cloud X UEs after data split. Therefore, when CA Performance Boost Based on Multi-dimensional Optimization takes effect, the benefits provided by intra-base-station CA downlink data split for Cloud X services decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-cell interference avoidance	INTER_CELL_INTRF_AVOID_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	On a given gNodeB, if CA Performance Boost Based on Multi-dimensional Optimization is enabled only for some of the cells for which inter-cell interference avoidance is enabled, the number of cells in which CA Performance Boost Based on Multi-dimensional Optimization takes effect may be less than the number of cells for which this function is enabled. Therefore, it is recommended that CA Performance Boost Based on Multi-dimensional Optimization be enabled for all cells under the gNodeB for which inter-cell interference avoidance is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink precise RB estimation	PRECISE_DL_RB_ESTIMATE_SW option of the NRDUCellDisch.AlgoSwitch parameter	<i>Downlink Scheduling</i>	When enabled together with downlink precise RB estimation, CA Performance Boost Based on Multi-dimensional Optimization may produce increased benefits.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station CA downlink data split	CLOUD_X_INTRA_GNB_CA_ENH_SW option of the NRDUCellAlgoSwitch.CloudVrSwitch parameter	<i>Layered QoS</i>	While CA Performance Boost Based on Multi-dimensional Optimization aims to improve UE scheduling efficiency and thereby increase the average downlink throughput of CA UEs, intra-base-station CA downlink data split aims to ensure the delay and user experience of R18 PDU Set UEs after data split. Therefore, when CA Performance Boost Based on Multi-dimensional Optimization takes effect, the benefits provided by intra-base-station CA downlink data split decrease.

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.3.4 Networking

There are at least two TDD frequencies deployed on the network.

6.3.5 Cells

CA Performance Boost Based on Multi-dimensional Optimization:

- This function requires the cells involved to be served by the same gNodeB, and it can take effect in a maximum of 18 cells under a gNodeB.
- This function also requires the cells involved to have the same frame offset configuration. The frame offset specified by the **gNBFreqBandConfig.FrameOffset** parameter preferentially takes effect. If this parameter is not configured or is set to **4294967295**, the frame offset specified by the **gNodeBParam.FrameOffset** parameter takes effect instead.

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

[Table 6-2](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 6-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
CA Enhanced Algorithm Extended Switch	NRDUCellCarrMgmt. <i>CaEnhancedAlgoExtSwitch</i>	MULTI_FREQ_CONVERGENT_SCH_SW	You are advised to select this option.

Parameter Name	Parameter ID	Option	Setting Notes
DL Scheduling Switch	gNodeBAlgo.DLSchSwitch	CA_CONCURRENT_SCHED_CAPB_INCR_SW	You are advised to select this option.
CA Enhanced Algorithm Extended Switch	NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch	CA_PFRM_BOOST_BSD_ON_MD_OPT_SW	You are advised to select this option. Before CA Performance Boost Based on Multi-dimensional Optimization is enabled, it is recommended that downlink fast intra-gNodeB CA (controlled by the CA_ULTRA_LOW_LATENCY_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) and CA concurrent scheduling capability increase (controlled by the CA_CONCURRENT_SCHED_CAPB_INCR_SW option of the gNodeBAlgo.DLSchSwitch parameter) be enabled.

6.4.1.2 Using MML Commands

Before using MML commands, refer to [6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Enable the multi-frequency joint scheduling function for both the PCell and SCells. The following uses the configuration for one cell as an example.

```
//Enabling multi-frequency joint scheduling
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoExtSwitch=MULTI_FREQ_CONVERGENT_SCH_SW-1&CA_PFRM_BOOST_BSD_ON_MD_OPT_S
```

```
W-1;  
MOD GNODEBALGO: DISchSwitch=CA_CONCURRENT_SCH_CAPB_INCR_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Disable the multi-frequency joint scheduling function for both the PCell and SCells. The following uses the configuration for one cell as an example.

```
//Disabling multi-frequency joint scheduling  
MOD NRDUCELLCARRMGMT: NrDuCellId=0,  
CaEnhancedAlgoExtSwitch=MULTI_FREQ_CONVERGENT_SCH_SW-0&CA_PFRM_BOOST_BSD_ON_MD_OPT_S  
W-0;  
MOD GNODEBALGO: DISchSwitch=CA_CONCURRENT_SCH_CAPB_INCR_SW-0;
```

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

After multi-frequency convergent scheduling is enabled, it has taken effect if any of the following changes is observed:

- The **User Downlink Average Throughput (DU)** increases.
- The proportion of traffic in SCells, which is the result of $\frac{N.CA.SCell.ThpVol.DL.Cell}{(N.CA.SCell.ThpVol.DL.Cell + N.CA.PCell.ThpVol.DL.Cell)}$, increases.

After CA Performance Boost Based on Multi-dimensional Optimization is enabled, it has taken effect if the average downlink throughput of CA UEs increases. The average downlink throughput of CA UEs is calculated using the following formula:
$$\frac{(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot)/N.CA.ThpTime.DL.RmvLastSlot.}$$

6.4.3 Network Monitoring

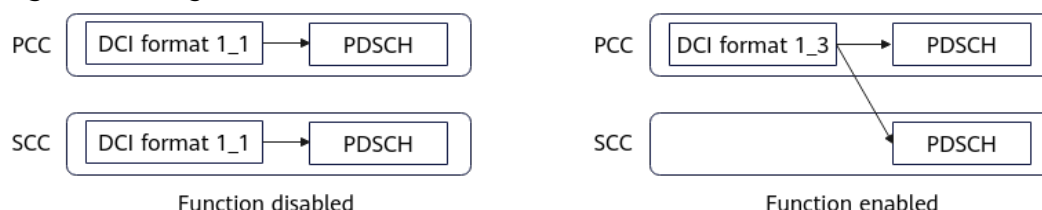
- Observe the average downlink UE throughput to check whether the UE data rate increases on the entire network after this function is enabled.
The average downlink UE throughput is indicated by **User Downlink Average Throughput (DU)**.
- Observe the average downlink throughput of CA UEs to check whether the CA UE data rate increases after this function is enabled.
The average downlink throughput of CA UEs is equal to the result of
$$(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot)/N.CA.ThpTime.DL.RmvLastSlot.}$$

7 Single DCI (Trial)

7.1 Principles

In multi-carrier scenarios, when a CA UE is scheduled on more than one carrier, carrier-specific CCE resources are used for indicating PDSCH scheduling on each of the carriers involved. The PDCCH load and PDSCH load vary with cells, and there may be idle PDSCH resources on a carrier while the PDCCH load of this carrier is heavy. In this case, the PDSCH on this carrier cannot be scheduled due to insufficient PDCCH CCEs. As a result, the average downlink throughput of CA UEs is negatively affected. Against this backdrop, the single DCI function is introduced. When the PDCCH CCEs on a carrier are insufficient, this function enables an inter-frequency carrier with sufficient CCEs to use DCI format 1_3 to schedule the PDSCHs on both carriers, as illustrated in [Figure 7-1](#). (In this scenario, the cell on the carrier with sufficient CCEs can be termed as a scheduling-dominant cell and the cell on the carrier with insufficient CCEs can be termed as a scheduling-subordinate cell.) This way, CCE consumption in the scheduling-subordinate cell decreases and the average downlink throughput of UEs increases. The single DCI function is controlled by the **DL_SINGLE_DCI_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter.

Figure 7-1 Single DCI function



- Before the single DCI function is enabled, each CC uses DCI format 1_1 to schedule the carrier-specific PDSCH. When the PDCCH load on the SCC is heavy, the downlink CCE allocation failure rate is high. Due to the downlink CCE allocation failures, the PDSCH on the SCC cannot be scheduled for CA UEs, affecting downlink experience of CA UEs.
- After the single DCI function is enabled, the PCC uses DCI format 1_3 to schedule the PDSCHs on both the PCC and SCC when the PDCCH load of the

PCC is light. (The PCell serves as the scheduling-dominant cell and the SCell as the scheduling-subordinate cell.) This scheduling method effectively relieves the PDCCH load of the SCC, reduces the number of PDSCH scheduling failures caused by downlink CCE allocation failures, and improves downlink experience of CA UEs.

 NOTE

- For details about DCI formats, see section 7.3.1.2 "DCI formats for scheduling of PDSCH" in 3GPP TS 38.212 V18.3.0.
- This function is applicable only to two intra-base-station carriers (one as the PCC and the other as the SCC) of the same duplex mode. In addition, this function requires that the PCell serve as the scheduling-dominant cell and the SCell serve as the scheduling-subordinate cell.

Single DCI is a trial function.

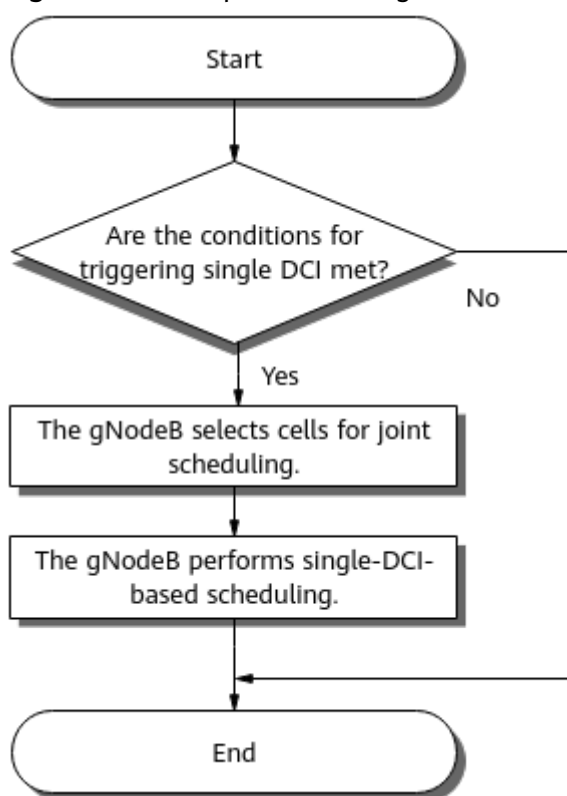
 NOTE

Trial functions are functions that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these functions can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial functions shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial functions. Trial functions are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial functions may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial functions but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial functions are free, Huawei is not liable for any trial function malfunctions or any losses incurred by using the trial functions. Huawei does not promise that problems with trial functions will be resolved in the current version. Huawei reserves the rights to convert trial functions into commercial functions in later R/C versions. If trial functions are converted into commercial functions in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial functions. If a customer fails to purchase such a license, the trial function(s) will be invalidated automatically when the product is upgraded.

Figure 7-2 shows the basic process of the single DCI function.

Figure 7-2 Basic process of single DCI



1. The conditions for triggering and exiting the single DCI function are described in [7.1.1 Triggering Conditions](#).
2. How the gNodeB selects cells for joint scheduling is described in [7.1.2 Selecting Cells for Joint Scheduling](#).
3. How the gNodeB performs single-DCI-based joint scheduling is described in [7.1.3 Performing Single-DCI-based Scheduling](#).

7.1.1 Triggering Conditions

If the single DCI function is enabled in the cells serving a CA UE, the gNodeB determines whether the CA UE can enter or exit the multi-carrier joint scheduling state based on cell conditions.

Entering Conditions

The gNodeB enables the CA UE to enter the multi-carrier joint scheduling state through RRC reconfiguration when the following cell conditions are met:

- The cells selected for joint scheduling meet the following configuration requirements:
 - The two cells have the same setting of the **NRDUCell.SubcarrierSpacing** parameter.
 - The two cells work in the same duplex mode, that is, the **NRDUCell.DuplexMode** parameter is set to **CELL_TDD** for both cells.
 - The two cells have the same setting of the **NRDUCellCsirs.TrsPeriod** parameter.

- The PCell and SCell are deployed on the same base station, that is, intra-base-station CA is used.
- In the case of NR TDD, the slot configurations of the PCell and SCell must meet the requirements listed in one scenario in the following table.

Table 7-1 Requirements for the slot configurations of the PCell and SCell

Slot Configuration Scenario	PCell Slot Configuration	SCell Slot Configuration
1	4:1 slot configuration (4_1_DDDSU)	4:1 slot configuration (4_1_DDDSU)
2	Single-period 8:2 slot configuration (8_2_DDDDDDDSUU)	Single-period 8:2 slot configuration (8_2_DDDDDDDSUU)

- The cells selected for joint scheduling meet the following dynamic conditions:
 - Scheduling-dominant cell:
 - Downlink CCE allocation failure rate < 5%
 - Uplink CCE allocation failure rate < 10%
 - Cell equivalent CCE usage < 25%
 - Scheduling-subordinate cell:
 - Downlink CCE allocation failure rate > SCC CCE allocation failure rate threshold for single DCI
(*NRDUCellPdcch.SglDciSccCceFailRateThld*)
 - Downlink PRB usage < 60%

 NOTE

- Downlink CCE allocation failure rate = $(N.CCE.DL.AllocReq.Num - N.CCE.DL.AggLvl1Num - N.CCE.DL.AggLvl2Num - N.CCE.DL.AggLvl4Num - N.CCE.DL.AggLvl8Num - N.CCE.DL.AggLvl16Num) / N.CCE.DL.AllocReq.Num$
- Uplink CCE allocation failure rate = $(N.CCE.UL.AllocReq.Num - N.CCE.UL.AggLvl1Num - N.CCE.UL.AggLvl2Num - N.CCE.UL.AggLvl4Num - N.CCE.UL.AggLvl8Num - N.CCE.UL.AggLvl16Num) / N.CCE.UL.AllocReq.Num$
- Cell equivalent CCE usage = $(N.CCE.ULDCI.Used.Equivalent + N.CCE.DLDCI.Used.Equivalent) / N.CCE.Avail.Equivalent$

Exiting Conditions

After a UE enters the multi-carrier joint scheduling state, the base station instructs the UE to exit the state through RRC reconfiguration if the cells for joint scheduling meet the relevant conditions.

- The scheduling-dominant cell meets any of the following conditions:
 - The single DCI function is disabled in this cell (which is the PCell).
 - The downlink CCE allocation failure rate is greater than 15%.
 - The uplink CCE allocation failure rate is greater than 30%.
- The scheduling-subordinate cell meets either of the following conditions:
 - The single DCI function is disabled in this cell (which is an SCell).
 - The downlink CCE allocation failure rate is less than 10%, and the cell equivalent CCE usage is less than 20%.

 NOTE

If the setting of the single DCI switch of the scheduling-dominant cell is changed from on to off while the single DCI switch of the scheduling-subordinate cell is kept on, the scheduling-dominant cell will no longer process the joint scheduling requests from the scheduling-subordinate cell.

7.1.2 Selecting Cells for Joint Scheduling

Scheduling-dominant Cell

Only the PCell can serve as the scheduling-dominant cell.

Scheduling-subordinate Cell

Single DCI allows only two cells for joint scheduling. If multiple SCells of a UE meet the conditions for multi-carrier joint scheduling, the base station selects one of the SCells based on their cell bandwidths and downlink CCE allocation failure rates.

- In terms of the cell bandwidth, the SCell with a large downlink bandwidth is preferentially selected.
- In terms of the downlink CCE allocation failure rate, the SCell with a high CCE allocation failure rate is preferentially selected.

The base station first checks the downlink cell bandwidths and then the CCE allocation failure rates. If the cells have the same downlink bandwidth, the base station checks the downlink CCE allocation failure rates of the cells. If the cells also have the same downlink CCE allocation failure rate, the base station selects the cell with the smallest **sCellIndex** value in the *SCellConfig* IE of the RRC_RECFG message for joint scheduling.

7.1.3 Performing Single-DCI-based Scheduling

If single DCI has taken effect for a CA UE, the scheduling-dominant cell uses DCI format 1_3 to carry the PDSCH scheduling information of both the scheduling-dominant cell and scheduling-subordinate cell.

If single DCI takes effect for an excessive number of UEs, the downlink experience of background UEs in the PCell will be affected. To prevent this issue, the operator can use the **NRDUCellPdcch.SingleDciAvailCceRatio** parameter to specify an appropriate percentage of CCEs available for single-DCI UEs.

 NOTE

- If the setting of the switch controlling the single DCI function is changed, the new setting takes effect only for UEs that access the network afterwards.
- After the single DCI function takes effect for a UE, the scheduling of the UE in the scheduling-subordinate cell depends on the scheduling-dominant cell. If the setting of the switch controlling the single DCI function is changed from selected to deselected for either the scheduling-dominant or scheduling-subordinate cell, the UE can be scheduled only in the scheduling-dominant cell.

7.2 Network Analysis

7.2.1 Benefits

On an NR TDD network, it is recommended that single DCI be enabled when PDCCH CCE resources are severely imbalanced between the two intra-base-station carriers. This function can increase the **User Downlink Average Throughput (DU)**.

7.2.2 Impacts

The impacts between this function and other functions are described in [7.3.2.3 Function Impacts](#).

The single DCI function may have the following impacts on the network:

- The downlink CCE usage (calculated using the **N.CCE.Used.Avg** and **N.CCE.Avail.Avg** counters) of the PCell increases.
- The average downlink CCE aggregation level of the PCell increases.
$$\text{Average downlink CCE aggregation level} = (1 \times \text{N.CCE.DL.AggLvl1Num} + 2 \times \text{N.CCE.DL.AggLvl2Num} + 4 \times \text{N.CCE.DL.AggLvl4Num} + 8 \times \text{N.CCE.DL.AggLvl8Num} + 16 \times \text{N.CCE.DL.AggLvl16Num}) / (\text{N.CCE.DL.AggLvl1Num} + \text{N.CCE.DL.AggLvl2Num} + \text{N.CCE.DL.AggLvl4Num} + \text{N.CCE.DL.AggLvl8Num} + \text{N.CCE.DL.AggLvl16Num})$$
- After this feature is enabled, the CA- and non-CA-related traffic distributions in the cell involved change. As a result, the values of CA-related counters or indicators also change. For example, the values of **N.CA.PR.B.DL.PCell.Used.Avg** and **N.CA.PR.B.DL.SCell.Used.Avg**, and the average throughput of CA UEs with SCells activated will change. The average throughput of CA UEs with SCells activated is calculated using the following formula: $(\text{N.CA.SccActiveUser.ThpVol.DL} - \text{N.CA.ThpVol.DL.LastSlot}) / \text{N.CA.ThpTime.DL.RmvLastSlot}$.

7.3 Requirements

7.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-081234	Single DCI (Trial)	NR0S000SDC00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Single DCI can take effect only after intra-band CA is enabled.
Low-frequency TDD	Intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Single DCI can take effect only after intra-FR inter-band CA is enabled.
Low-frequency TDD	CSI-RS Rate Matching	CSIRS_RATE_MATCH_SW option of the NRDUCellPd sch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	Single DCI can be enabled only after CSI-RS rate matching is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA and DRX status collaboration	CA_AND_DRX_STATUS_C OLLAB_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmExtSwitch parameter	<i>Carrier Aggregation</i>	Single DCI can be enabled only after CA and DRX status collaboration is enabled.
Low-frequency TDD	CSI-RS 4P/8P Adaptive Switching Thld	NRDUCellCsirs4P8PAdaptiveSwitchingThld	<i>Channel Management</i>	Single DCI can be enabled only when this threshold parameter is set to 0 .
Low-frequency TDD	Configuration of the maximum port number used for CSI-RS for CM	NRDUCellCsirsFR1MaxCellCsirsPortNumber	<i>Channel Management</i>	Single DCI can be enabled only when this port number parameter is set to 8PORT .

7.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	High-speed cells (for which NRDUCell.HighSpeedFlag is set to HIGH_SPEED) do not support single DCI.
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Hyper cells do not support single DCI.
Low-frequency TDD	Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	If Fusion Cell has been enabled, single DCI cannot be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	- NRDUCell.NrDuCellNetworkMode set to HYPER_CELL_COMBINE_MODE - NRDUCellMultiTrp.DataTranMode set to BASIC_MODE or DPS_MODE	Cell Combination	Combined cells do not support single DCI.
Low-frequency TDD	Joint Transmission Cell Combination	- NRDUCell.NrDuCellNetworkMode set to HYPER_CELL_COMBINE_MODE - NRDUCellMultiTrp.DataTranMode set to JOINT_MODE	Cell Combination	Combined cells in joint transmission mode do not support single DCI.
Low-frequency TDD	CCE resource optimization	CCE_RESOURCE_OPT_SW option of the NRDUCellPdcch.PdcchAlgorithmSwitch parameter	MIMO (TDD)	Single DCI and CCE resource optimization are mutually exclusive. They cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH multi-symbol resource pool	MULTI_SYM_CORESET_SW option of the NRDUCellPdcch.PdcchAlgorithmSwitch parameter	<i>Channel Management</i>	Single DCI and PDCCH multi-symbol resource pool are mutually exclusive. They cannot be both enabled.
Low-frequency TDD	Configuration of multiple PDCCH search spaces	MULTI_SEARCH_SPACE_SW option of the NRDUCellPdcch.PdcchAlgorithmSwitch parameter	<i>Channel Management</i>	Single DCI and "configuration of multiple PDCCH search spaces" are mutually exclusive. They cannot be both enabled.
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgorithmSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	Single DCI and "low latency and high reliability" are mutually exclusive. They cannot be both enabled.
Low-frequency TDD	Downlink 8-layer transmission for a single FWA user	DL_SUEIGHT_LAYER_SW option of the NRDUCellDLRank.DlRankSelAlgorithmSw parameter	<i>FWA</i>	Single DCI and "downlink 8-layer transmission for a single FWA user" are mutually exclusive. They cannot be both enabled.
Low-frequency TDD	DC Component rate matching	DC_COMPONENT_RATE_MATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	Single DCI and DC component rate matching are mutually exclusive. They cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CCE partial symbol adaptation	CCE_PARTIAL_SYM_ADAPT_SW option of the NRDUCellPdcch.PdcchAllocationSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Single DCI and CCE partial symbol adaptation are mutually exclusive. They cannot be both enabled.
Low-frequency TDD	Downlink avoidance enhancement mode 2	NRDUCellTimeIntf.TimelntfAvoidance	None	Single DCI and downlink avoidance enhancement mode 2 are mutually exclusive. They cannot be both enabled.

7.3.2.3 Function Impacts

- Impact relationships with functions related to carrier management and MIMO are as follows.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CCE-based data split stop for CA UEs	CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	Single DCI takes precedence over CCE-based data split stop for CA UEs. If single DCI has taken effect for a UE, CCE-based data split stop for CA UEs will not take effect for this UE. If single DCI is triggered for a UE for which CCE-based data split stop for CA UEs has taken effect, CCE-based data split stop for CA UEs will be terminated for this UE.
Low-frequency TDD	Downlink fast inter-gNodeB CA	INTER_GNB_ULTRALOW_LAT_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>CA Experience Improvement</i>	Downlink fast inter-gNodeB CA does not take effect for 3CC CA UEs for which single DCI has taken effect.
Low-frequency TDD	Downlink inter-gNodeB CA	INTER_GNODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>CA Experience Improvement</i>	Downlink inter-gNodeB CA does not take effect for 3CC CA UEs for which single DCI has taken effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA SRS carrier switching	SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrierMgmt.CAEnhancedAlgorithmSwitch parameter	<i>CA Experience Improvement</i>	- When SRS carrier switching takes effect in a UE's SCell that is involved in joint scheduling, SRS power control adjustment does not take effect for the UE in this SCell. - For a small number of UEs that cannot be scheduled in the downlink N symbols before SRS transmission, the PDSCH may not be scheduled in self-contained slots of the scheduling-dominant cell because the number of start and length indicator value (SLIV) combinations is not enough.
Low-frequency TDD	Uplink intra-band CA	INTRABAND_UL_CASW option of the NRDUCellAlgorithmSwitch.CarrierAggregationSwitch parameter	<i>Carrier Aggregation</i>	If uplink intra-band CA has taken effect for a UE, single DCI will not take effect for the UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink intra-FR inter-band CA	INTRA_FR_INTER_BAND_UL_CASW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	If uplink intra-FR inter-band CA has taken effect for a UE, single DCI will not take effect for the UE.

- Impact relationships with other functions are as follows.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	If intra-base-station DL CoMP and multi-TRS measurement (controlled by the NRDUCellAlgoSwitch.MultiTrsMeasureSwitch parameter) are enabled in the PCell and a UE supports multi-TRS measurement, single DCI does not take effect for the UE.
Low-frequency TDD	Enhanced remote interference adaptive avoidance	RIM_ADAPT_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	Enhanced remote interference adaptive avoidance does not take effect for UEs for which single DCI has taken effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Clock out-of-synchronization detection	CLK_DETECT_SW option of the gNodeBParam.ClkOutofsyncDetSwitch parameter	<i>Synchronization</i>	Clock out-of-synchronization detection does not take effect for UEs for which single DCI has taken effect.
Low-frequency TDD	Downlink scheduling optimization during RRC reconfiguration	RRC_REC_CFG_DL_SCH_OPT_SW option of the NRDUCellDLISch.DLSchAlgoSwitch parameter	<i>Downlink Scheduling</i>	When downlink scheduling optimization during RRC reconfiguration takes effect, the downlink bit error rate of UEs for which single DCI has taken effect may slightly increase during RRC connection reconfiguration.
Low-frequency TDD	Multi-TRS measurement	NRDUCellAlgoSwitch.MultiTrsMeasurementSwitch	<i>High Speed Mobility</i>	If intra-base-station DL CoMP (controlled by the INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter) and multi-TRS measurement are enabled in the PCell and a UE supports multi-TRS measurement, single DCI does not take effect on the UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Adaptive fast processing by the gNodeB	GNB_ADAPT_FAST_PROCESS_SW option of the gNodeBParam.ServiceDiffProcSwitch parameter	None	If the single DCI function is enabled for any cell served by a gNodeB for which the "adaptive fast processing by the gNodeB" function is enabled, the latter function will not take effect on the gNodeB. In addition, scheduling on the SCC may fail while the setting of the single DCI function switch is being changed.
Low-frequency TDD	Downlink precise RB estimation	PRECISE_DL_RB_ESTIMATE_SW option of the NRDUCellDLISch.DLSchAgoSwitch parameter	<i>Downlink Scheduling</i>	When enabled together with downlink precise RB estimation, single DCI may produce increased benefits.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving BWP	BWP2_SWI TCH option of the NRDUCellUePwrSaving . BwpPwrSavingSw parameter	<i>UE Power Saving</i>	When both power saving BWP and single DCI are enabled on the PCC, whether DCI format 1_3 can be configured on the BWP2 of the PCC depends on the "single DCI configuration on BWP2" function (controlled by the BWP2_CONFIG_SINGLE_DCI_SW option of the NRDUCellCarrMgmt.CaCompatibilitySwitch parameter). • If "single DCI configuration on BWP2" is enabled, DCI format 1_3 can be configured on the BWP2 of the PCC. • If "single DCI configuration on BWP2" is disabled, DCI format 1_3 cannot be configured on the BWP2 of the PCC. If UEs have compatibility issues when both single DCI and power saving BWP are enabled, it is recommended that "single DCI

RAT	Function Name	Function Switch	Reference	Description
				configuration on BWP2" be enabled to reduce the downlink IBLER and RBLER of these UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Time-domain power saving BWP	TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	When both time-domain power saving BWP and single DCI are enabled on the PCC, whether DCI format 1_3 can be configured on the BWP2 of the PCC depends on the "single DCI configuration on BWP2" function (controlled by the BWP2_CONFIG_SINGLE_DCI_SW option of the NRDUCellCarrMgmt.CaCompatibilitySwitch parameter). • If "single DCI configuration on BWP2" is enabled, DCI format 1_3 can be configured on the BWP2 of the PCC. • If "single DCI configuration on BWP2" is disabled, DCI format 1_3 cannot be configured on the BWP2 of the PCC. If UEs have compatibility issues when both single DCI and time-domain power saving BWP take effect, it is

RAT	Function Name	Function Switch	Reference	Description
				recommended that "single DCI configuration on BWP2" be enabled to reduce the downlink IBLER and RBLER of these UEs.

7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.3.4 Networking

The network uses at least two frequencies.

7.3.5 Cells

Cells involved in joint scheduling must be deployed in the same BBU.

7.3.6 Others

UEs must support single DCI. A UE supports single DCI if the *nonSharedSpectrum-fdd-fr1* or *nonSharedSpectrum-tdd-fr1* IE with the value "supported" is present in the *coScheduledCellSCS-r18* parameter under *multiCell-PDSCH-DCI-1-3-SameSCS-r18* in the **CA-ParametersNR** field of the UECapabilityInformation message over the Uu interface. For details, see section 4.2.7.4 "CA-ParametersNR" in 3GPP TS 38.306 V18.2.0.

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

[Table 7-2](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
CA Enhanced Algorithm Switch	NRDUCellCarrMgmt. <i>CaEnhancedAlgoSwitch</i>	DL_SINGLE_DCI_SW	You are advised to select this option.
Single DCI SCC CCE Alloc Fail Rate Thld	NRDUCellPdcch.<i>SglDciSccCceFailRateThld</i>	N/A	Use the recommended value.
Single DCI Available CCE Ratio	NRDUCellPdcch.<i>SglDciAvailCceRatio</i>	N/A	Use the recommended value.
CA Compatibility Switch	NRDUCellCarrMgmt. <i>CaCompatibilitySwitch</i>	BWP2_CONFIG_SINGLE_DCI_SW	If UEs have compatibility issues when both single DCI and power saving BWP are enabled, it is recommended that this option be selected.

7.4.1.2 Using MML Commands

Before using MML commands, refer to [7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Enable the single DCI function for both the PCell and SCell. The following uses the configuration for one cell as an example.

```
//Enabling single DCI
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=DL_SINGLE_DCI_SW-1;
MOD NRDUCELLPDCCH: NrDuCellId=0, SingleDciAvailCceRatio=60, SglDciSccCceFailRateThld=20;
//(Optional) Enabling "single DCI configuration on BWP2" if UEs have compatibility issues when both single
DCI and power saving BWP are enabled
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaCompatibilitySwitch=BWP2_CONFIG_SINGLE_DCI_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Disable the single DCI function for both the PCell and SCell. The following uses the configuration for one cell as an example.

```
//Disabling single DCI
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=DL_SINGLE_DCI_SW-0;
//(Optional) Disabling "single DCI configuration on BWP2"
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaCompatibilitySwitch=BWP2_CONFIG_SINGLE_DCI_SW-0;
```

7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.2 Activation Verification

- Step 1** Start Uu signaling tracing on the MAE as follows: Log in to the MAE and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.
- Step 2** Enable a UE to access the network. Check the fields in the *ServingCellConfig* IE of the *rrcReconfiguration* message delivered by the gNodeB for SCell configuration. If the *ServingCellConfig* IE contains the **mc-DCI-SetOfCellsToAddModList-r18** field, the single DCI function has taken effect.

----End

7.4.3 Network Monitoring

- Observe the average downlink UE throughput (**User Downlink Average Throughput (DU)**) to check the UE data rate increase on the entire network after this function is enabled.
- Observe the average downlink throughput of CA UEs to check the CA UE data rate increase after this function is enabled. The average downlink throughput of CA UEs is equal to $(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot) / N.CA.ThpTime.DL.RmvLastSlot$.

8 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

 NOTE

You can find the EXCEL files of parameter reference for the software version on the live network from the product documentation delivered with that version.

FAQ: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-020100.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

9 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

10 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

11 Reference Documents

- 3GPP TS 38.306: "NR; User Equipment (UE) radio access capabilities"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.212: "NR; Multiplexing and channel coding"
- *EPC-based NSA Basics*
- *Carrier Aggregation*
- *CA Experience Improvement*
- *Multi-Operator Sharing*
- *High Speed Mobility*
- *Hyper Cell*
- *Cell Combination*
- *Channel Management*
- *Uplink Scheduling*
- *Energy Conservation and Emission Reduction*
- *Multi-Frequency Smart Selection*
- *Co-MM (Low-Frequency TDD)*
- *NR Inter-Carrier Dynamic Power Allocation*
- *Ultimate Cloud X Experience*
- *License Control Item Lists*
- *CoMP*
- *MIMO (TDD)*
- *UE Power Saving*
- *Massive MIMO AHR (Low-Frequency TDD)*
- *Multi-RAT Coordinated Energy Saving*
- *Remote Interference Management (Low-Frequency TDD)*
- *URLLC*
- *Synchronization*
- *Downlink Scheduling*
- *Fusion Cell (Low-Frequency TDD)*

- *FWA*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- *Layered QoS*